Internal Assignment : July 2026

Program Name M.Sc (Mathematics)

Paper Code – MAT-101 (Advanced Algebra)

Section-A
(Very short Answer Type question)

Q 1

(i) If $O(G)=4$ and $O(H)=5$ than find $O(G\times H)$

Sol(i)

$$O(G\times H) = O(G)\cdot O(H)$$
$$= 4\times 5$$
$$\Rightarrow O(G\times H)=20$$

Note!- $O(G)$ denoter the order of group G, which is number of elements in G.

(ii) If $t: V\to V'$ is the zero homomorphism then find $im(t)$

Sol(ii)

$\because$ $t: V\to V'$

s.t $t(x)=0$ $\quad x\in V$ and $0\in V'$

where 0 is the zero element of V'

then $im(t) = \{0\}$

(iii) Define unique factorization domain.

Sol^{n}(iii) Let R be an integral domain. Then R is said to be a unique factorization domain (UFD) if any non-zero element of R is either a unit or it can be expressed as the product of a finite number of prime element and this product is unique up to associates. Thus, if $a\in R$ is a non-zero, non unit element, then

(i) $a = x_1 x_2 \cdots x_m$, x_i $(1 \le i \le m)$ are prime in R

(ii) If $a = y_1 y_2 \cdots y_n$ also, where y_j $(1 \le i \le n)$ are prime in R, then $m = n$ and each x_i is an associate of some x_i and vice-versa

(IV) Define transcendental extension of a field

Soln (IV) Let K be a field extension of a field F. If every element of K is algebric over F then an extension K over F is called algebric extension of F. If K is not an algebric extension of F, it is called a transcendental extension of F

Section-B

(Short Answer Question)

Note :- Answer any two question. Each answer should be given in 200 words.

2 If G be a group and $N \triangle G$. If N and G/N are solvable then prove that G is solvable.

Soln Let us suppose that N and G/N are solvable, Then there exists a slovable series for N as

$$N = N_0 \supset N_1 \supset N_2 \supset \cdots N_K = \{e\}$$

Also a solvable series for G/N as

$$\frac{G}{N} = \frac{G_0}{N} \supset \frac{G_1}{N} \supset \frac{G_2}{N} \supset \cdots \supset \frac{G_m}{N} = \{N\}$$

such that $N_{i+1} \triangleleft N_i$, N_i/N_{i+1} is abelian for $i = 0, 1, 2 \cdots K-1$ and

$$\frac{G_{i+1}}{N} \triangleleft \frac{G_i}{N}, \frac{(G_i/N)}{(G_{i+1}/N)}$$

is abelian for $j = 0, 1, 2 \ldots m-1$

From series (2), we observe that each G_i is a subgroup of G and N is a normal subgroup of each G_i

Now, since $\frac{G_{j+1}}{N} \triangleleft \frac{G_i}{N}$, therefore $G_{j+1} \triangleleft G_j$, for all $j = 0, 1, 2 \ldots m-1$

Also $\frac{G_m}{N} = \{N\} \Rightarrow G_m = N$

Then $G = G_0 \supset G_1 \supset G_2 \supset \ldots G_m = N = N_0 \supset N_1 \supset N_2 \supset \ldots N_k = \{e\}$

is a solvable series for G

Hence G is solvable.

3. If R be a Euclidean ring, then prove that every non zero element in R is either a unit or can be written as product of a finite number of prime element of R

Soln - Let a be a non-zero element of R. If a is unit, then the theorem is true. Now, if a is prime, then the theorem is again true. Let us suppose that a is neither a unit nor a prime in R. Let d be the corresponding Euclidean valuation of R, we shall prove the theorem by mathematical induction on $d(a)$. Let us suppose that the theorem is true for all element $r \in R$ for which $d(r) < d(a)$ Then we shall prove the theorem for a

which is a non-unit and reducible element of R since a is reducible, therefore

$$a = xy$$

where neither x nor y is a unit in R

Now x, y are not unit in R, then by theorem 8, we have

$$d(x) < d(xy) \quad \text{and} \quad d(y) < d(xy)$$
$$d(x) < d(a) \quad \text{and} \quad d(y) < d(a)$$

Thus, by induction assumption x and y can be written as a product of finite number of prime element of R. Let us suppose that

$$x = x_1 x_2 \cdots x_n \quad \& \quad y = y_1 y_2 \cdots y_m$$

where $x_i (i = 1, 2, \cdots n)$ and $y_i (j = 1, 2, \cdots m)$ are prime element of R. Then by (i)

$$a = x_1 x_2 \cdots x_n \cdot y_1 y_2 \cdots y_m$$

that is, a has been expressed as a product of finite number of prime element of R.

Section-C
(Long Answer Question)

7 If V is an inner product space then prove that for any vector $u, v \in V$

(a) $\|u+v\|^2 - \|u-v\|^2 = 4(u,v)$

(b) $\|u+v\|^2 + \|u-v\|^2 = 2(\|u\|^2 + \|v\|^2)$

Soln

Proof:- We have $\|u+v\|^2 = \langle u+v, u+v\rangle$

$\Rightarrow \langle u,u\rangle + \langle u,v\rangle + \langle v,u\rangle + \langle v,v\rangle$

$\Rightarrow \langle u,u\rangle + 2\langle u,v\rangle + \langle v,v\rangle \quad [\because \langle u,v\rangle = \langle v,u\rangle]$

$= \|u\|^2 + 2\langle u,v\rangle + \|v\|^2$

$\|u+v\|^2 = \langle u-v, u-v\rangle$

$= \langle u,u\rangle + \langle u,-v\rangle + \langle -v,u\rangle + \langle -v,-v\rangle$

$= \langle u,u\rangle - \langle u,v\rangle - \langle v,u\rangle + (-1)^2 \langle v,v\rangle$

$= \|u\|^2 - 2\langle u,v\rangle + \|v\|^2$

From (1) and (2) we get

$\|u+v\|^2 - \|u-v\|^2 = 4\langle u,v\rangle$ and

$\|u+v\|^2 + \|u-v\|^2 = 2(\|u\|^2 + \|v\|^2)$

Hence the theorem